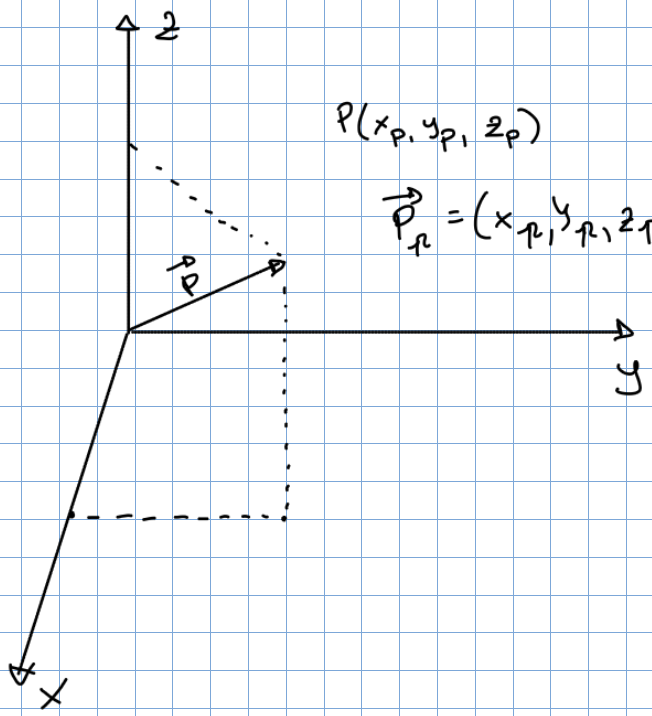
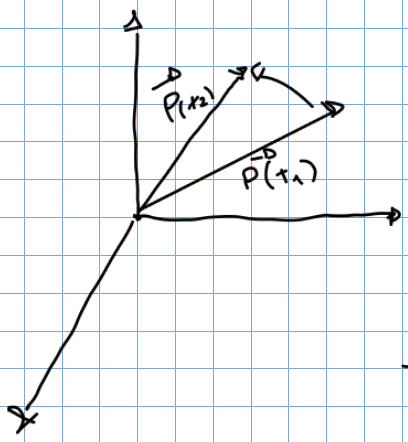




$\Delta\vec{R} = \vec{p}(t_2) - \vec{p}(t_1)$

$\Delta\vec{R} \neq 0$

è di spostamento

$\Delta t = t_2 - t_1$

$\Delta\vec{R} = \vec{p}(t_2) - \vec{p}(t_1)$

$\vec{v}_m = \dfrac{\Delta\vec{R}}{\Delta t} = \dfrac{\vec{p}(t_2) - \vec{p}(t_1)}{t_2 - t_1}$

↳ velocità media

$$\lim_{t_2 \to t_1} \frac{\vec{p}(t_2) - \vec{p}(t_1)}{t_2 - t_1} = \frac{d\vec{p}}{dt} = \vec{v}$$

= velocità istantanea

$(v_x, v_y, v_z) = \vec{v}(t)$

Moto rettilineo uniforme

$\vec{v}$ = costante $\Rightarrow \begin{cases} v_x = \text{cost} \\ v_y = \text{cost} \\ v_z = \text{cost} \end{cases}$

Il problema della cinetica è: $\vec{r}(t) = (x(t), y(t), z(t))$

$\vec{v} = \vec{v}_0 = (v_{0x}, v_{0y}, v_{0z})$

visto che $\vec{v} = \dfrac{d\vec{r}}{dt}$ allora

$v_x = \dfrac{dx}{dt}$, $v_y = \dfrac{dy}{dt}$, $v_z = \dfrac{dz}{dt}$

quindi

$v_{x_0} = \dfrac{dx}{dt}$, $v_{y_0} = \dfrac{dy}{dt}$, $v_{z_0} = \dfrac{dz}{dt}$

$\dfrac{df}{dt} = k \Rightarrow df = k\,dt \Rightarrow \displaystyle\int_{f(t_1)}^{f(t_2)} df = \int_{t_1}^{t_2} k\,dt \Rightarrow f(t_2) - f(t_1) = k(t_2 - t_1)$

$\Downarrow$

$f(t_2) = f(t_1) + k(t_2 - t_1)$

$\Downarrow$

dalle quali ricaviamo

$\displaystyle\int_{f(t_1)}^{f(t_2)} df = f(t_2) - f(t_1)$

$\displaystyle\int_{t_1}^{t_2} k\,dt = k\int_{t_1}^{t_2} dt = k(t_2 - t_1)$

$v_{x_0} = \dfrac{dx}{dt} = x(t_2) = x(t_1) + v_{0x}(t_2 - t_1) \overset{t_1 = t_0 \,\&\, t_2 = t}{\Rightarrow} x(t) = x_0 + v_x(t_2 - t_0)$

$v_{y_0} = \dfrac{dy}{dt} \Rightarrow y(t) = y_0 + v_y(t_2 - t_0)$

$v_{z_0} = \dfrac{dz}{dt} \Rightarrow z(t) = z_0 + v_z(t_2 - t_0)$

$$\Rightarrow \vec{P}(t) = \vec{P}_0 + \vec{v}_0 (t - t_0)$$

$$\Delta \vec{r} = \vec{r}(t) - \vec{r}_0 = \vec{v}_0 (t - t_0) \Rightarrow \frac{\Delta \vec{R}}{\Delta t} = \vec{v}_0 \qquad \frac{dx}{dx} = 1$$

$$\vec{v} = \frac{d}{dt}\left(\vec{r} + \vec{v}_0 (t - t_0)\right) = \frac{d\vec{B}}{dt} + \frac{d}{dt}\left[\vec{v}_0 (t - t_0)\right] = 0 + \vec{v}_0 \frac{d}{dt}(t - t_0)$$

$$= \vec{v}_0 \left(\frac{dt}{dt} - \frac{dt_0}{dt}\right) = \vec{v}_0$$

vettore accelerazione

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2 \vec{r}}{dt^2} \rightarrow \text{moto uniformemente accelerato}$$

$$\vec{a} = \vec{a}$$

$$\vec{a}_0 = (a_{0x}, a_{0y}, a_{0z})$$

$$a_{0x} = \frac{dv_x}{dt} \Rightarrow dv_x = a_{0x}\, dt \Rightarrow \int_{v_x(t_1)}^{v_x(t_2)} dv_x = \int_{t_1}^{t_2} a_{0x}\, dt = \int_{v_1(t_0)}^{v_1(t)} dv_x = \int_{t_0}^{t} a_{0x}\, dt$$

$$= v_x(t) - v_x(t_0)$$

$$= a_{0x}(t - t_0) \Rightarrow v_x(t) = v_x(t_0) + a_{0x}(t - t_0)$$

$$a_{0y} = \frac{dv_y}{dt} \Rightarrow \text{comp}$$

$$a_{0z} = \frac{dv_z}{dt} \Rightarrow \text{comp}$$

$$\Rightarrow v_x(t) = v_{0x} + a_{0x}(t - t_0)$$

$$v_x = \frac{dx}{dt} = v_{0x} + a_{0x}(t - t_0)$$

$$\Rightarrow dx = v_x(t)\, dt = \left[v_{0x} + a_{0x}(t - t_0)\right] dt$$

$$\Rightarrow \int_{x(t_0)}^{x(t)} dx = \int_{t_0}^{t} \left[v_{0x} + a_{0x}(t - t_0)\right] dt \Rightarrow x(t) - x(t_0) + \int_{t_0}^{t} dt\, a_{0x}(t - t_0) =$$

$$= \int_{t_0}^{t} a_{0x}(t - t_0)\, dt \longrightarrow \int_0^{t-t_0} \text{eS}\; a_{0x} S = a_{0x} \int_0^{t-t_0} dS - S =$$

$$= a_{0x} \frac{S^2}{2} \Big|_0^{t-t_0} = \frac{a_{0x}}{2}(t-t_0)^2$$

$$S = t - t_0 \Rightarrow \begin{cases} t = t_0 \Rightarrow S = 0 \\ t = t \Rightarrow S = t - t \end{cases}$$

Mi sono preso $\frac{1}{1}$